

# Gayathri Emuru

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[Portfolio](#)

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## Summary

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- Data Scientist with **4+ years** of experience across industry and academic research solving high-impact problems using **machine learning, AI, NLP, and real-world data**.
- Skilled in Python, SQL, experimentation, predictive modeling, and building end-to-end data and AI solutions from data preparation to deployment and monitoring.
- Proven ability to turn complex data into measurable business impact through scalable systems, clear insights, and strong collaboration with business and technical teams.
- [AWS Certified Machine Learning Engineer](#) – Associate with hands-on experience in building machine learning and NLP solutions.

## Work Experience

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### Data Science Research Assistant - University of Delaware

#### SADL Lab - Data Science & NLP

May 2025 – present

- Published a [research paper](#) detailing sophisticated data extraction and classification techniques for CITES participants achieving 100% validated accuracy in data extraction, accompanied by detailed data visualizations using Power BI to identify complex data patterns.
- Authored a peer-reviewed [research paper](#) on a multimodal NLP pipeline that processed 35,000+ political speeches and 192,000+ images using transformer embeddings, BERTopic, and CLIP, achieving 83% topic classification accuracy, and created multilingual, geocoded corpora for LLM benchmarking, retrieval, and political forecasting; dataset publicly released on [Harvard Dataverse](#).
- Developed an **AI-driven bilingual NLP pipeline** combining FAISS retrieval, DeBERTa-v3-large NLI, LLM-assisted text comparison, and Bayesian changepoint modeling to detect, validate, and explain selective content withheld from English-language audiences.

#### Jayaraman Research Group - Hyperspectral ML

Jan 2025 – May 2025

- Architected an [end-to-end Python ML pipeline](#) for ToF-SIMS hyperspectral analysis, transforming high-dimensional ion spectra into pixel-level chemical segmentation maps at 65,536-pixel scale using SOM, PCA, t-SNE, and UMAP - directly supporting 3M materials R&D decisions.

#### Lenhoff Research Group - Computer Vision

Aug 2024 – Jan 2025

- Developed a **15-class image classification** model using an ensemble of ResNet50, DenseNet121, and EfficientNetBo for membrane 3D structural data, achieving 99% accuracy; implemented Grad-CAM for model interpretability.
- Developed a **Python-based ETL pipeline** to transform 162 inconsistent text and image PDFs into structured CSV files with 95% accuracy, employing advanced data cleaning methods to maintain data integrity.

### Data Scientist | Nagarro

July 2021 – Aug 2024

- Built an **XGBoost model** on AWS Sage Maker to classify stocks under Additional Surveillance Measures (ASM), achieving 83% accuracy and improving DVM score reliability.
- Developed an automated stock portfolio analysis system with interactive dashboards, leveraging advanced clustering techniques and data visualization to drive a **37% increase in monthly active users**.
- Led Agile development of financial web pages, integrating real-time stock feeds, statistical analysis, earnings call podcasts, business intelligence insights, and email OTPs.
- Implemented **MLOps best practices** with Git for version control and developed **186 REST APIs to handle 700M** monthly hits, optimizing data mining for 18 large B2B stockbroker clients.
- Developed and deployed **computer vision algorithms** (YOLOv3, OCR-PyTesseract) with TensorFlow-Keras to digitize handwritten purchase orders, achieving 92% accuracy and improving operational efficiency with Docker-based deployment.
- Designed and implemented a payments module for Africell in Angola, improving transaction speeds by 30% and ensuring 99.9% accuracy for 10M users using NumPy and Pandas for data processing.

Education

|                        |                                                   |            |                      |
|------------------------|---------------------------------------------------|------------|----------------------|
| University of Delaware | Master of Science in Data Science                 | CGPA: 4/4  | Aug 2024 - May 2026  |
| MTIET (A.P - India)    | Bachelor of Technology(B.Tech) - Computer Science | GPA:8.5/10 | Aug 2017 - June 2021 |

Skills

- **Programming Languages:** Python, SQL, C.
- **Data Analysis & Machine Learning:** Pandas, NumPy, EDA - Data visualization (Matplotlib, Seaborn, Power BI, Tableau), Classical Machine Learning, Deep Learning, NLP, Computer Vision, Statistical methods, XGBoost, LightGBM, AWS, System Design, Scikit-Learn, Generative Models (GMM, GAN's), Image Recognition, Forecasting, Sentiment Analytics, Optimization Techniques, Feature Engineering, Generative AI, LLM.
- **Data Science Processes:** A/Btesting, ETL, Data Cleaning, Time Series, Recommender Systems.
- **Software & Tools:** Django (Full Stack), MySQL, Git, GitHub Actions, CI/CD Pipelines, JIRA, REST APIs, Beautiful Soup, Big Data (PySpark), Excel, Linux, TensorFlow, PyTorch, MLFlow, Streamlit, Jupyter Notebooks, R Studio.

Projects [[GitHub](#)]

- **Generate Anime Images using GAN's: Applied Generative AI(GenAI) - GANs** to create high fidelity Anime images, optimizing model performance during training.
- **County-Level Mental Health Prediction:** Built an end-to-end Python pipeline by integrating multi-source U.S. public health and socioeconomic datasets across 3,100+ counties and 30+ variables, performing feature engineering, imputation, EDA, and visualization, and training Random Forest and Decision Tree models that achieved  $R^2 = 0.71$ ,  $F1 = 0.90$ , and  $AUC = 0.77$ .
- **Signature Verification:** Implemented deep learning techniques (Computer Vision), specifically **Siamese CNN**, for Signature Verification, achieving a significant 73% accuracy in differentiating genuine and forged signatures.
- **Educational Project Approval Prediction** for DonorsChoose.org: Predicted proposal approvals using NLP Techniques and Machine Learning Algorithms achieving a test accuracy of 0.66.